

Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

10.01 Interpreting graphs (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: Gold Coast
2. Ex 1b: Uluru
3. Ex 2a: 15–24 years
4. Ex 2b: 12%
5. Ex 3a: Jan and Feb, 18°C
6. Ex 3b: it falls steadily to a low in July

Easy · Set A

1. Gold Coast
2. Uluru
3. 30
4. true (8)

Easy · Set A2 (quick drill)

1. 15
2. 50
3. 28
4. 12
5. 25
6. 20

Easy · Set B

1. 15–24 years
2. 12%
3. 16.5%
4. e.g. working-age adults and migrants

Medium · Set C

1. Jan and Feb, 18°C
2. e.g. Jan and Feb (or May and Sep)
3. October
4. July (3°C)
5. May–June or June–July (2°C)
6. it falls steadily to a low in July

Challenging · Set D

1. 22%
2. Anglican
3. No religion
4. e.g. more Christians and fewer “no religion”

Extension

1. 76.5°
2. Gold Coast 6 cm, Uluru 3.4 cm, Snowy 3 cm, Kangaroo Is. 2 cm, Phillip Is. 1.6 cm

Exit ticket

1. Level 1: Phillip Island
2. Level 2: 11°C
3. Level 3: 8%

10.02 Misleading graphs (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: the vertical axis starts at \$2 million, not 0
2. Ex 1b: profits grew much faster than they did
3. Ex 2a: no
4. Ex 2b: 51 and 56
5. Ex 3a: uneven scale, pictures, no labels, missing data
6. Ex 3b: to impress investors

Easy · Set A

1. axis does not start at 0
2. huge growth
3. \$2.2 million
4. \$1.8 million

Easy · Set A2 (quick drill)

1. M
2. F
3. M
4. M
5. M
6. F

Easy · Set B

1. 50
2. 51
3. 58
4. no

Medium · Set C

1. line graph from 0 to 4 with even scale

Challenging · Set D

1. the area is 4 times bigger, not 2
2. you do not know what is shown
3. the steps are not equal
4. own answer

Extension

1. the axis starts at 2, so heights above 2 are compared (0.2 vs 2.0)
2. e.g. axis from 0 to 20

Exit ticket

1. Level 1: e.g. the axis does not start at 0
2. Level 2: small differences look huge
3. Level 3: start the vertical axis at 0 with an even scale

10.03 Dot plots (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 23°C
2. Ex 1b: 15°C
3. Ex 2a: 4
4. Ex 2b: $\frac{1}{15}$
5. Ex 3a: dot plot
6. Ex 3b: most days between 20°C and 26°C, peak at 23°C

Easy · Set A

1. 23°C
2. 15°C
3. 4
4. $\frac{1}{15}$

Easy · Set A2 (quick drill)

1. 3
2. 8
3. 2
4. 12
5. 7
6. 3

Easy · Set B

1. 16
2. 1
3. 7
4. 8

Medium · Set C

1. dot plot
2. dot plot

Challenging · Set D

1. dot plot
2. mode 0, outlier 15; it makes the mean bigger

Extension

1. 18.8%
2. still 1

Exit ticket

1. Level 1: a graph with one dot for each value on a number line
2. Level 2: 1
3. Level 3: 8

10.04 Stem-and-leaf plots (booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: 25
2. Ex 1b: 71 kg
3. Ex 2a: 80 kg
4. Ex 2b: 106 kg
5. Ex 3a: 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
6. Ex 3b: to see the shape and find the median and mode easily

Easy · Set A

1. 25
2. 71 kg
3. 80 kg
4. 106 kg

Easy · Set A2 (quick drill)

1. 27
2. 50
3. 19
4. 83
5. 104
6. 66

Easy · Set B

1. 15
2. 45
3. 75
4. 66

Medium · Set C

1. 2 | 1 3 7 8; 3 | 0 2 5 6 9; 4 | 1 4 5
2. 6 | 4 7 8; 7 | 0 3 5 8 9; 8 | 2 5 8; 9 | 1

Challenging · Set D

1. 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
2. 9 players, 9/25

Extension

1. 10 | 5; 11 | 2 8; 12 | 1 5; 13 | 0
2. 25 hundredths (0.25)

Exit ticket

1. Level 1: 6
2. Level 2: 40, 42, 47
3. Level 3: 1 | 2 4 7; 2 | 1 5 9

10.05 Mean and mode (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 4
2. Ex 1b: 4
3. Ex 2a: 8.78
4. Ex 2b: 6
5. Ex 3a: 3.5; 1
6. Ex 3b: 6

Easy · Set A (quick drill)

1. 4
2. 6
3. 1
4. 7
5. 12
6. red
7. 4
8. 3
9. 5
10. 20
11. 5
12. 5

Easy · Set B

1. 5
2. 4
3. 30
4. 5
5. 3.5
6. 9

Medium · Set C

1. 8.78; 6
2. 3.5; 1
3. 14; 15
4. 4.07; 3.5
5. 49.86; 52
6. 1.88; 2

Challenging · Set D

1. 6
2. 84
3. e.g. 8, 8, 10, 11, 13
4. yes, e.g. 1, 1, 2, 2 (modes 1 and 2)

Extension

1. 10.4
2. it is a balance point, e.g. mean of 1 and 2 is 1.5

Exit ticket

1. Level 1: 5
2. Level 2: 8
3. Level 3: 6.5

10.06 Median and range (booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: 4
2. Ex 1b: 4
3. Ex 2a: 10
4. Ex 2b: 6
5. Ex 3a: 3.5; 7
6. Ex 3b: the mean

Easy · Set A (quick drill)

1. 4
2. 6
3. 7
4. 19
5. 13
6. 8
7. 3
8. 5
9. 7
10. 3
11. 30
12. 6.5

Easy · Set B

1. 8
2. 4
3. 5
4. 7
5. 6
6. 22.5

Medium · Set C

1. 10; 6
2. 3.5; 7
3. 14.5; 6
4. 3.85; 1.6
5. 52; 23
6. 4; 98

Challenging · Set D

1. mean 7, median 3.5, mode 3, range 23
2. the median: the outlier 25 pulls the mean up
3. e.g. 1, 4, 6, 8, 11
4. 57

Extension

1. 3, 5, 9, 9
2. a few very high salaries make the mean misleading

Exit ticket

1. Level 1: 6
2. Level 2: 25
3. Level 3: 9

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 39)**WE DO examples (teacher)**

1. Ex 1a: 7
2. Ex 1b: 1.5
3. Ex 2a: 66
4. Ex 2b: 64
5. Ex 3a: 61.2
6. Ex 3b: mostly in the 60s

Easy · Set A

1. 7
2. 1.5
3. 1
4. 1.88

Easy · Set A2 (quick drill)

1. 6
2. 16
3. 0
4. 20
5. 8
6. 18

Easy · Set B

1. 30
2. 64
3. 66
4. 61.2

Medium · Set C

1. 13
2. 23
3. 23
4. 23.2
5. 23.48
6. the range

Challenging · Set D

1. 35
2. 83
3. 80
4. 85.88

Extension

1. the median: it is not affected by extreme values
2. 1.53

Exit ticket

1. Level 1: 3
2. Level 2: 53.5
3. Level 3: 18

10.08 Comparing data (booklet p. 45)**WE DO examples (teacher)**

1. Ex 1a: the marks of the girls (left) and boys (right)
2. Ex 1b: 94, a girl
3. Ex 2a: 54
4. Ex 2b: 83
5. Ex 3a: girls: mean 55, median 54; boys: mean 55.8, median 60
6. Ex 3b: girls: range 76 vs boys 83

Easy · Set A

1. 94, girl
2. 10, boy
3. 8
4. 13

Easy · Set A2 (quick drill)

1. A
2. B
3. A
4. B

Easy · Set B

1. 55
2. 54
3. 76
4. 55.8
5. 60
6. 83

Medium · Set C

1. Class A: 5 | 8 2, 4 | 5 ...; Class B on the right
2. Class A: higher median, smaller range

Challenging · Set D

1. boys have a slightly higher median (60 vs 54) but a larger range
2. girls: smaller range (76 vs 83)
3. median, as there are some very low marks
4. own questions

Extension

1. Ana: she scores 12 or more every game
2. e.g. 5, 5, 5 and 0, 5, 10

Exit ticket

1. Level 1: comparing two groups
2. Level 2: A
3. Level 3: A